

JOURNEY: HARNESS ARCHITECTURE DEFENSE

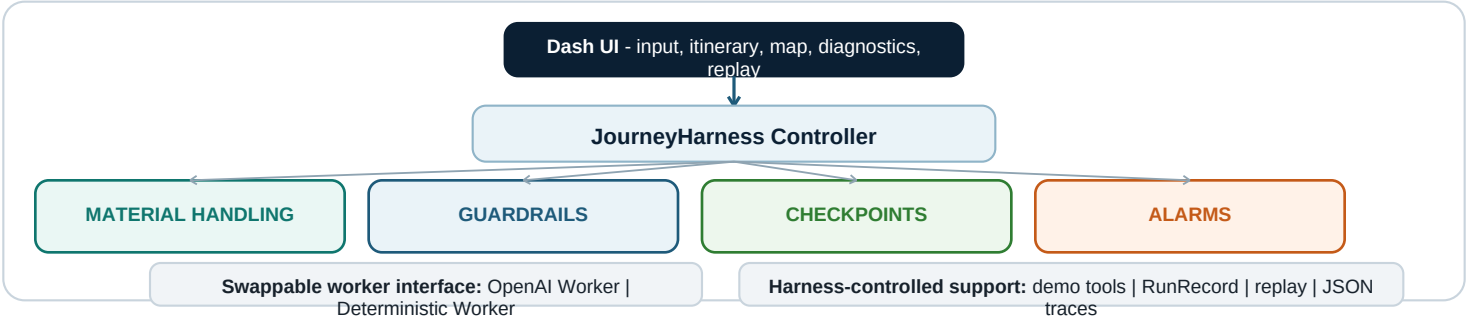
Hamzah Ahmed | 24-Hour Build Challenge

Harness-first | Fixture-only data | Swappable workers

GOAL

Journey is a reusable harness that governs an itinerary-planning worker. A user enters a **real trip request** for the supported destination. The harness packages local demo place data, constrains the worker, evaluates its draft, requests one bounded repair, and stops for human input when it cannot safely continue.

ARCHITECTURE



FOUR REQUIRED PILLARS

1 GUARDRAILS

Declared rules validate trip input, restrict actions, cap candidates, and enforce time, token, refinement, and repair limits. A rule may **block**, **transform**, or **escalate**; every result is recorded.

2 CHECKPOINTS

Explicit **pass/fail criteria** evaluate input integrity, candidate quality, itinerary schema, place grounding, and schedule feasibility. A checkpoint returns feedback; it never silently edits worker output.

3 MATERIAL HANDLING

Versioned Pydantic envelopes carry trip input, fixture places, ranked candidates, worker context, drafts, checkpoint feedback, and validated plans. Each material has a source, stage, hash, and schema version.

4 ALARMS

Structured failures contain a named type, severity, stage, context, recommended action, and human-review flag. Alarms are persisted, logged, and shown in diagnostics.

CHECKPOINT FEEDBACK CHANGES WORKER BEHAVIOR

Unknown place ID "place-999" -> Grounding checkpoint fails -> **GROUNDING_VIOLATION** alarm -> harness returns valid IDs and a required correction -> same worker repairs once -> checkpoints run again -> **pass and persist**, or **REPAIR_EXHAUSTED + HUMAN_REVIEW_REQUIRED**.

WORKER AND TOOLS

Workers: OpenAI and deterministic workers implement the same protocol. **Tools:** load the supported destination, filter local fixture places, rank deterministically, and create a save payload. The harness calls tools; the worker receives only ranked IDs and prepared context.

HUMAN-IN-THE-LOOP

The harness stops and asks rather than guesses when no candidates match, constraints conflict, a repair fails twice, or a time/token/refinement limit is reached.

PERSISTENCE AND REPLAY

A typed **RunRecord** stores materials, guardrail and checkpoint results, alarms, worker attempts, metrics, and status in browser-local storage. Restored records are revalidated. Replay starts after a selected checkpoint and reuses earlier material instead of repeating completed stages.

DEMO SCOPE AND TRADEOFFS

One supported destination, 10-15 bundled places, one- to three-day trips, and clearly labeled simulated ratings and costs. This favors reliability, repeatable tests, checkpoint replay, and a five-minute demo over live place coverage. OpenAI is used only when selected; the deterministic worker provides a no-key fallback and proves portability.